

EDUCATION

• McMaster University

September 2024 - April 2029 (Expected)

Computer Engineering (B.Eng) - CO-OP | Dean's List

Hamilton, Ontario

- **Relevant Coursework:** Principles of Programming, Data Structures & Algorithms, Microprocessor Systems, Logic Design, Circuit Analysis, Operating Systems

PROJECTS

• Jarvis: AI Voice Assistant

January 2025 - Present

Tools: Python, LangChain, Ollama, Whisper, Kokoro TTS, PySide6, Porcupine, Playwright, SQLite



- Engineered a **fully local, privacy-first** AI voice assistant with a Central Dispatcher orchestrating **8+ autonomous agents** (calendar, weather, news, search, smart home, timer, tasks, browser) via a **plugin architecture** requiring zero core changes to extend.
- Implemented a **multi-threaded audio pipeline**: Porcupine wake-word (~50 ms), real-time Whisper STT, and **sentence-buffered Kokoro TTS streaming**, reducing perceived response latency by ~40%.
- Built a **FunctionGemma 270M router** classifying queries in ~50 ms across **15+ functions** with **VRAM lifecycle management** and auto-unload timeouts for GPU memory optimization.
- Designed a **PySide6 Fluent Design GUI** with lazy-loading tabs, streaming chat, and a **VLM browser agent** (Playwright + Qwen3-VL) for autonomous web navigation.

• 3D Spatial Mapping System

February 2025 - April 2025

Tools: C, Python, ARM Cortex-M4F (MSP432E401Y), VL53L1X ToF, I2C, UART, matplotlib



- Built an embedded **indoor 3D spatial mapper** using a rotating IR ToF sensor on a stepper motor, producing interactive wireframe models applicable to **autonomous navigation and LiDAR prototyping**.
- Captured **640 data points** across 10 scans at **64 pts/scan** with **5.625° resolution**, achieving a theoretical quantization error of **0.061 mm** per reading.
- Designed a real-time pipeline: I2C at **100 kbps** → on-chip storage → UART at **115,200 bps** → Python **64-bit** polar-to-Cartesian conversion, eliminating 32-bit FPU rounding error.
- Implemented interrupt-driven controls (falling-edge ISRs), alternating CW/CCW motor direction, and coil de-energization and more.

• Personal Portfolio (AI-Powered)

December 2026 - January 2026

Tools: TypeScript, Next.js, React, TailwindCSS, Sanity CMS, OpenAI API, Clerk Auth



- Designed and deployed a next-gen portfolio featuring an **AI Twin chatbot** powered by OpenAI, enabling visitors to interact with a conversational clone trained on personal and professional context.
- Integrated **Sanity headless CMS** for dynamic content management and **Clerk authentication**; built a fully responsive UI with **dynamic routing** optimized for desktop and mobile.

EXPERIENCE

• Toronto District School Board

September 2023 - June 2024

Coding Instructor

Etobicoke, Ontario

- Taught **20+ students** Java, Python, and Scratch, simplifying complex programming concepts for diverse learners.
- Designed lesson plans and debugged student code in real time, fostering problem-solving and teamwork.
- Introduced real-world software development concepts such as modular design and reusability; provided individualized technical support.

SKILLS

- **Programming Languages:** Python, C/C++, Java, HTML/CSS, JavaScript, TypeScript, MATLAB, LaTeX, XML
- **Databases & Backend:** AWS, Azure, Supabase, Firebase, MongoDB, SQLite
- **Frameworks/Libraries:** Next.js, React, TailwindCSS, LangChain, Whisper, Ollama, numPy, OpenCV, YOLO, PySide6
- **Hardware:** Verilog, Arduino, ARM Cortex-M4F, Quanser Technologies, Microcontrollers
- **Tools:** GitHub, Git, Figma, VS Code, MATLAB, Microsoft Office, PowerBI, Keil MDK,
- **Extras:** National Lifeguard, Air Cadets, French Diploma